End-to-End Workflow

Based on LR-FHSS Framework and 3GPP TR 38.811

Cross-Layer Pipeline

- Orbit -> satellite position
- Coverage -> user distribution
- Load -> active devices
- Channel -> SNR / SINR
- Decoding -> successful packets
- Power -> energy consumption

Project Repository

https://github.com/SatNaingTun/LRFHSS_MultiBeam_Integration.git

Problem Statement

What Is Estimated at Each Step

Given satellite state at step t, estimate:

1. covered population $P_{\mathrm{cov}}(t)$
2. calculated nodes $N_{\mathrm{node}}(t)$
3. calculated demodulators $N_{\mathrm{demod}}(t)$
4. elevation-conditioned load and decoding behavior
5. busy/idle/sleep demod split and power

Workflow

- Set input data, simulation steps, and elevation scenarios.
- Move the satellite step by step along the orbit.
- For each step, estimate covered population.
- Convert covered population into active nodes and available demodulators.
- For each elevation case, split demodulators into busy, idle, and sleep states.
- Run LR-FHSS decoding simulation for each elevation case.
- Save output files for each step.
- Merge all steps into time-series results.
- Plot time vs decoded payloads.
- Plot time vs energy.
- Plot time vs demodulator states.

Orbit Mean Motion

$$n = \sqrt{\frac{\mu}{a^3}}$$

- Derivation: from circular/two-body orbit dynamics, equating gravitational acceleration μ/a^2 to centripetal acceleration an^2 gives $n^2 = \mu/a^3$.
- Symbols: n mean motion, μ Earth gravitational parameter, a semi-major axis.
- Ref: Orbital mechanics (Vallado) https://doi.org/10.1007/978-1-4939-0802-8

Mean Anomaly Update

$$M(t) = M_0 + n(t - t_0)$$

- Derivation: mean anomaly advances linearly in time with angular speed n, so integrating $\frac{dM}{dt} = n$ gives $M(t) = M_0 + n(t - t_0)$.
- Symbols: $M(t)$ anomaly at time t, M_0 anomaly at reference epoch t_0, n mean motion.
- Ref: Orbital mechanics (Vallado) https://doi.org/10.1007/978-1-4939-0802-8

Horizon Central Angle

$$\psi_h = \arccos\left(\frac{R_E}{r_{\mathrm{orb}}}\right)$$

- Derivation: at the Earth-limb tangent point, radius and line-of-sight are perpendicular, forming a right triangle with adjacent/hypotenuse ratio R_E/r_{orb}.
- Symbols: ψ_h horizon central angle, R_E Earth radius, r_{orb} orbital radius.
- Ref: NTN geometry context https://www.3gpp.org/DynaReport/38.811.htm

Geometric Footprint Radius

$$R_{\text{geo}} = R_E \psi_h$$

- Derivation: surface coverage radius is modeled as arc length on Earth, and arc length equals $R_E \times$ central angle.
- Symbols: R_{geo} geometric footprint radius, R_E Earth radius, ψ_h horizon central angle.
- Ref: NTN geometry context https://www.3gpp.org/DynaReport/38.811.htm

Node Mapping

$$N_{\text{node}}(t) = P_{\text{pop}}\rho_{\text{node}}$$

- Derivation: proportional scaling assumption; expected active nodes are modeled as a fixed penetration ratio of covered population.
- Symbols: $N_{\text{node}}(t)$ calculated nodes, P_{pop} population base, ρ_{node} node/population ratio.
- Ref: Implementation rule in `modules/satellite_stepper.py`

Demod Mapping

$$N_{\text{demod}}(t) = P_{\text{pop}}\rho_{\text{demod}}$$

- Derivation: proportional capacity sizing; demodulator count is modeled as a fixed ratio of covered population.
- Config values (current defaults): $\rho_{\text{node}} = 10^{-7}$, $\rho_{\text{demod}} = 10^{-7}$.
- Symbols: $N_{\text{demod}}(t)$ calculated demodulators, P_{pop} population base, ρ_{demod} demod/population ratio.
- Ref: Implementation rule in `modules/satellite_stepper.py`

Elevation-Dependent Link Pressure

Implementation-Faithful Model

The implementation uses elevation-dependent distance and a relative FSPL pressure term:

$$F_e(t) = \left(\frac{d_e(t)}{d_{\text{ref}}}\right)^2, \qquad d_{\text{ref}} = \frac{H_{\text{sat}}}{1000}\ \text{km}$$

- Derivation: Friis distance dependence implies received power scales as $1/d^2$; using a zenith reference normalizes this into a relative load-pressure factor.
- $d_e(t)$: mean user distance at elevation case e.
- d_{ref}: zenith-reference distance.
- $F_e(t)$: relative path-loss pressure used for load scaling.
- Ref: implemented in `modules/satellite_stepper.py`

Elevation-Dependent Offered Load

From Distance to Demodulator Demand

$$A_e(t) = \max(1, N_e(t))\, \alpha_{\text{act}}\, F_e(t)$$

- Derivation: offered load is modeled as active users scaled by activity ratio and link-pressure factor.

$$B_e(t) = \text{ErlangB}(N_{\text{demod}}(t), A_e(t))$$

- Derivation: blocking probability follows Erlang-B loss system with servers $N_{\text{demod}}(t)$ and offered load $A_e(t)$.

$$N_{\text{busy},e}(t) = \text{round}(\min\left(N_{\text{demod}}(t), A_e(t)\left(1 - B_e(t)\right)\right))$$

- Derivation: carried load is $A_e(1 - B_e)$; busy demodulators are the capped, integer version of carried load.

Elevation Effect on Energy

State Transition Chain in Implementation

$$N_{\mathrm{rem},e}(t) = N_{\mathrm{demod}}(t) - N_{\mathrm{busy},e}(t)$$

- Derivation: demodulators are partitioned by conservation of count into busy + non-busy.

$$N_{\mathrm{sleep},e}(t) = \mathrm{round}(\beta_{\mathrm{sleep}} N_{\mathrm{rem},e}(t)) \,, \quad N_{\mathrm{idle},e}(t) = N_{\mathrm{rem},e}(t) - N_{\mathrm{sleep},e}(t)$$

- Derivation: remaining demodulators are split by a fixed sleep ratio; idle is the residual to preserve total count.
- Lower elevation usually increases $d_e(t)$.
- Larger $d_e(t)$ increases $F_e(t)$ and then $A_e(t)$.
- Higher carried load increases busy demodulators and modeled power.

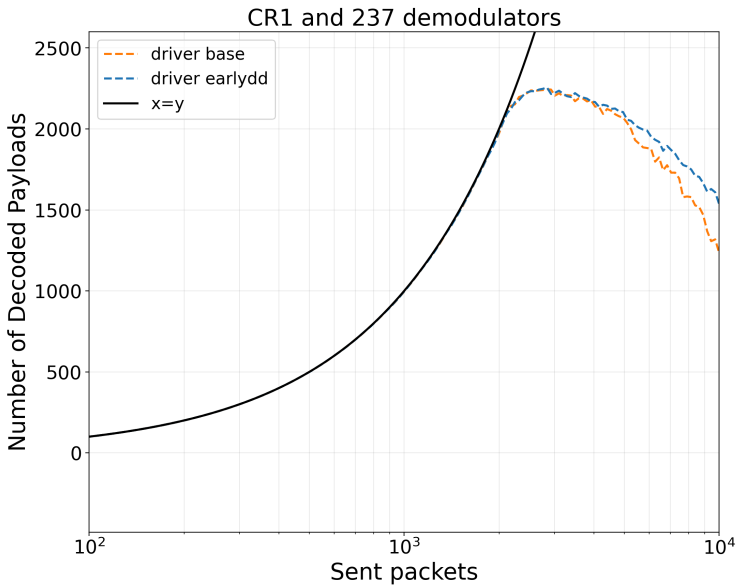
Power Model

$$P_e(t) = P_0 + N_{\text{idle},e}(t)P_{\text{idle}} + N_{\text{busy},e}(t)P_{\text{busy}}$$

- Derivation: linear state-additive power model, where each demod state contributes its per-unit power plus baseline platform power.
- Symbols: $P_e(t)$ total modeled power for elevation scenario e.
- Symbols: P_0 baseline power, P_{idle} idle demod power, P_{busy} busy demod power.
- Symbols: $N_{\text{idle},e}(t)$ and $N_{\text{busy},e}(t)$ are stepwise demod state counts.
- Current constants: $P_0 = 2$ mW, $P_{\text{idle}} = 9$ mW, $P_{\text{busy}} = 100$ mW.
- Ref: power-state modeling basis https://tnm.engin.umich.edu/wp-content/uploads/sites/353/2017/12/2006.10.Reducing-idle-mode-power-in-software-defined-radio-terminals_ISLPED-2006.pdf

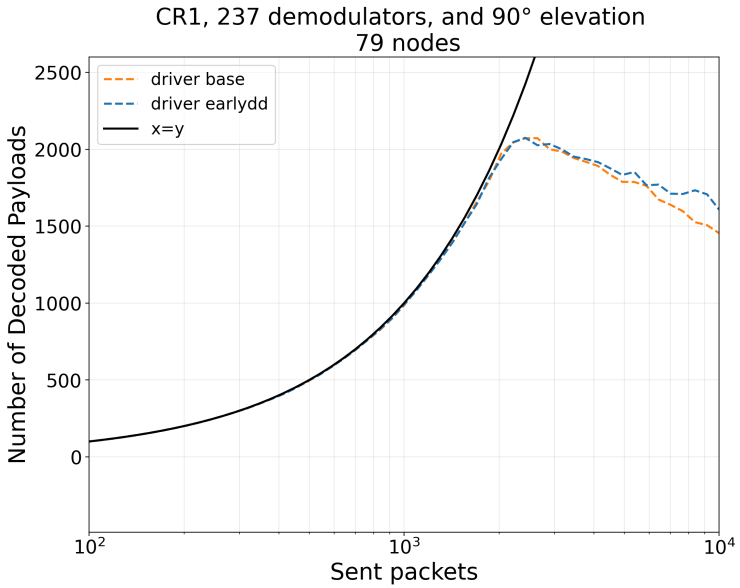
One-Position Decode Results

Not Fixed Elevation



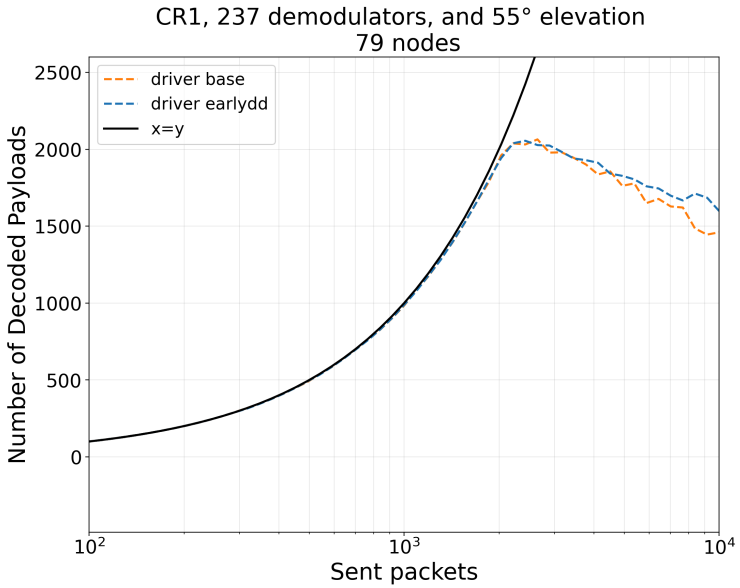
One-Position Decode Results

Elevation Curves (90 deg)



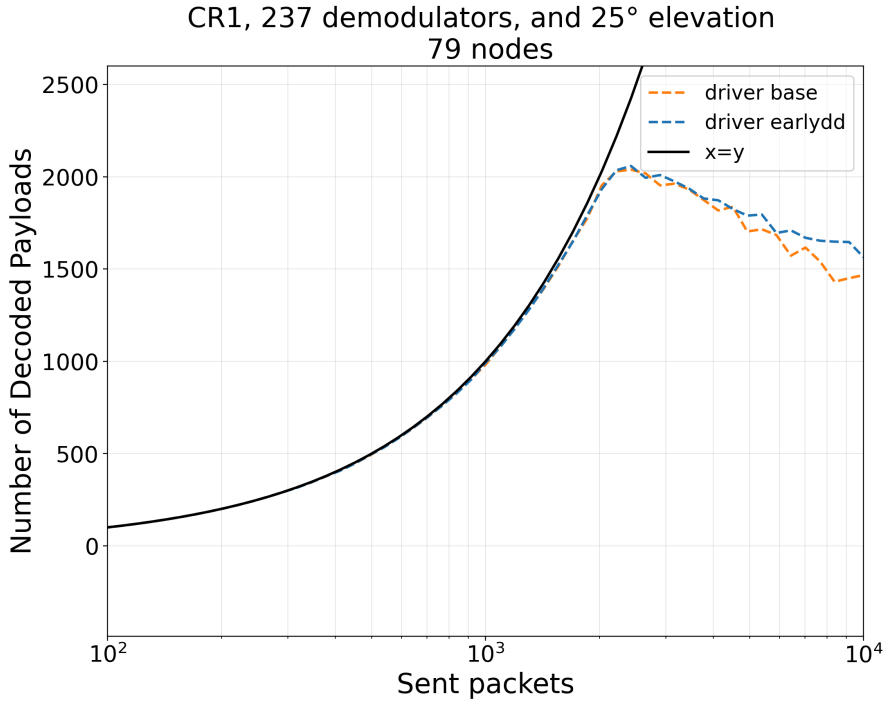
One-Position Decode Results

Elevation Curves (55 deg)



One-Position Decode Results

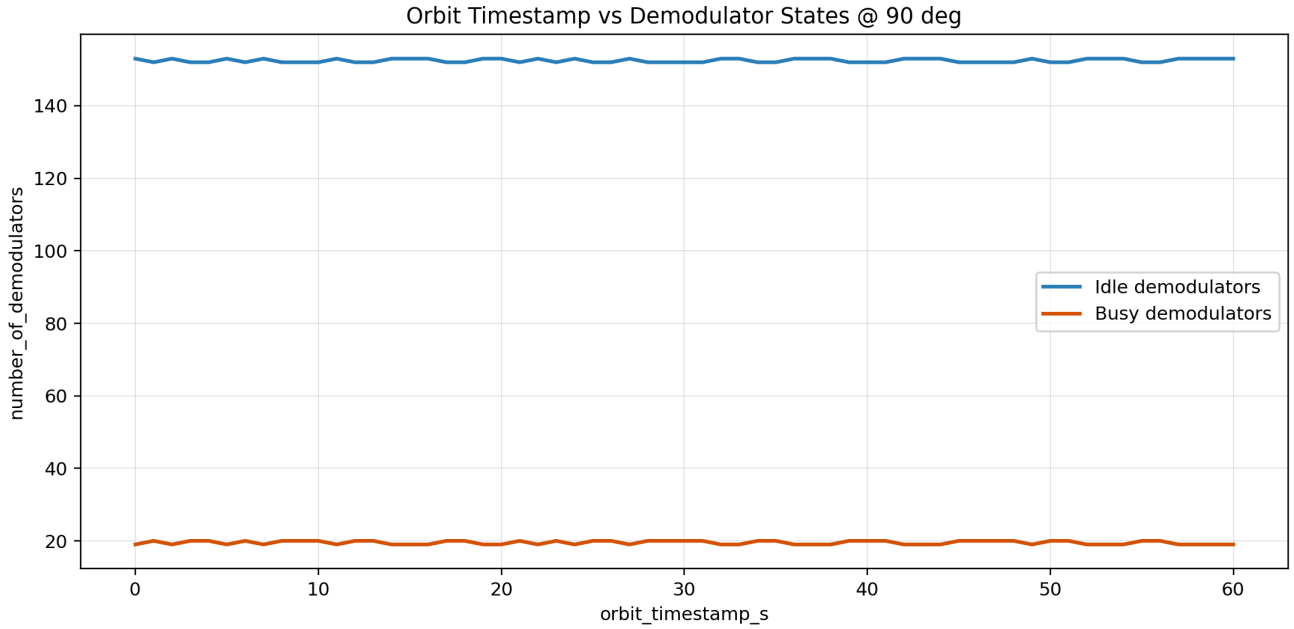
Elevation Curve (25 deg)



- Same demod budget, stronger geometry penalty at lower elevation.

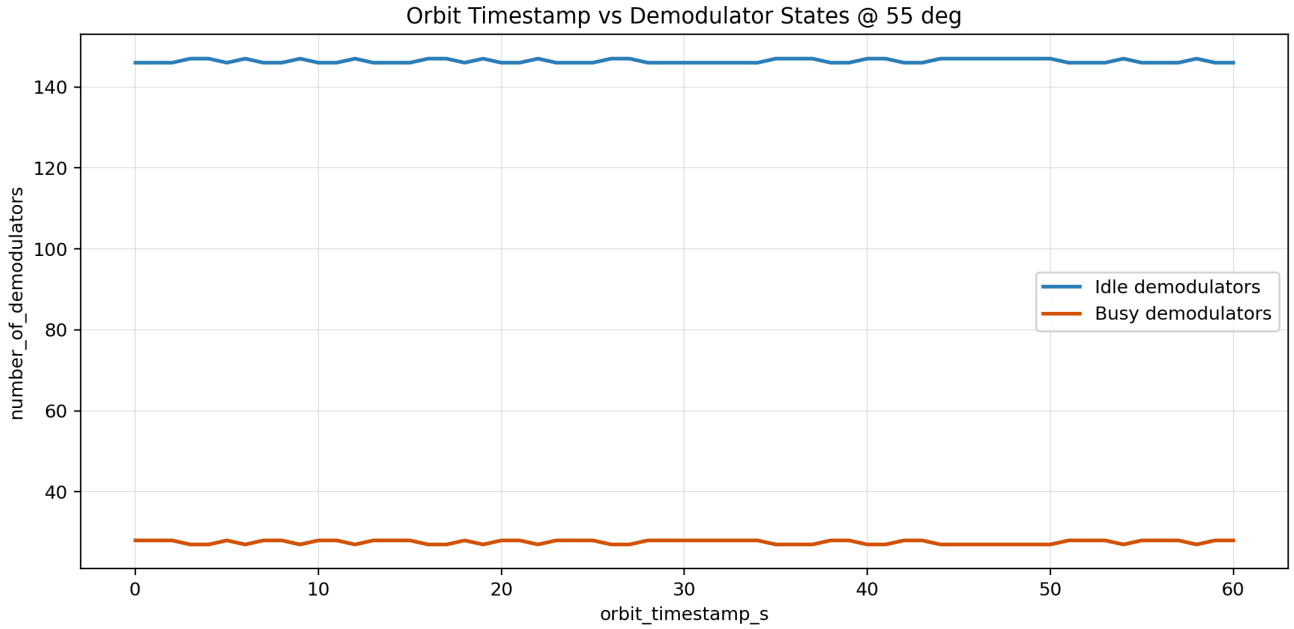
Demodulator State Evidence

Busy/Idle vs Orbit Timestamp (90 deg)



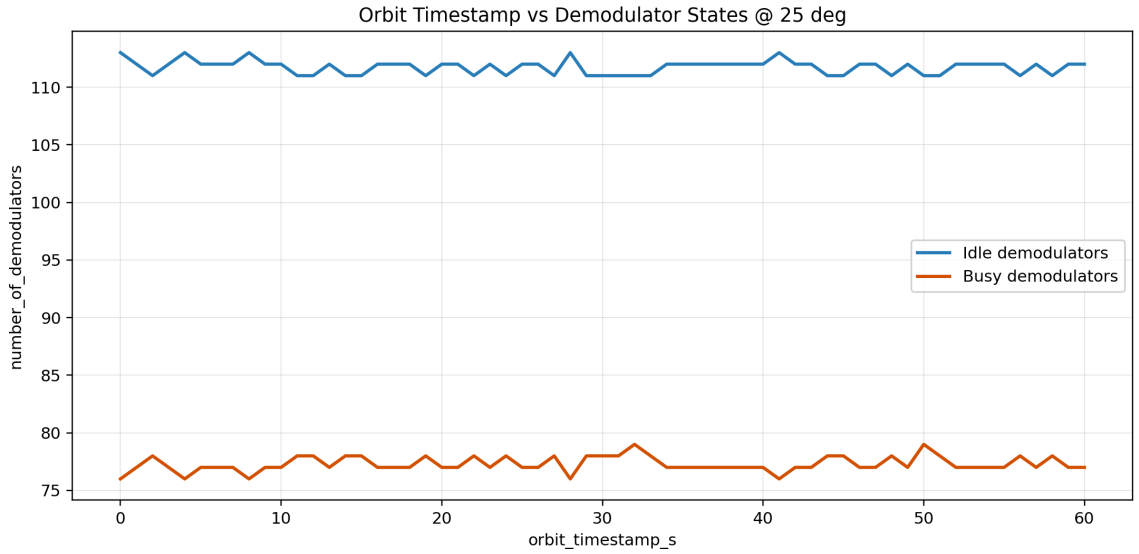
Demodulator State Results

Busy/Idle vs Orbit Timestamp (55 deg)

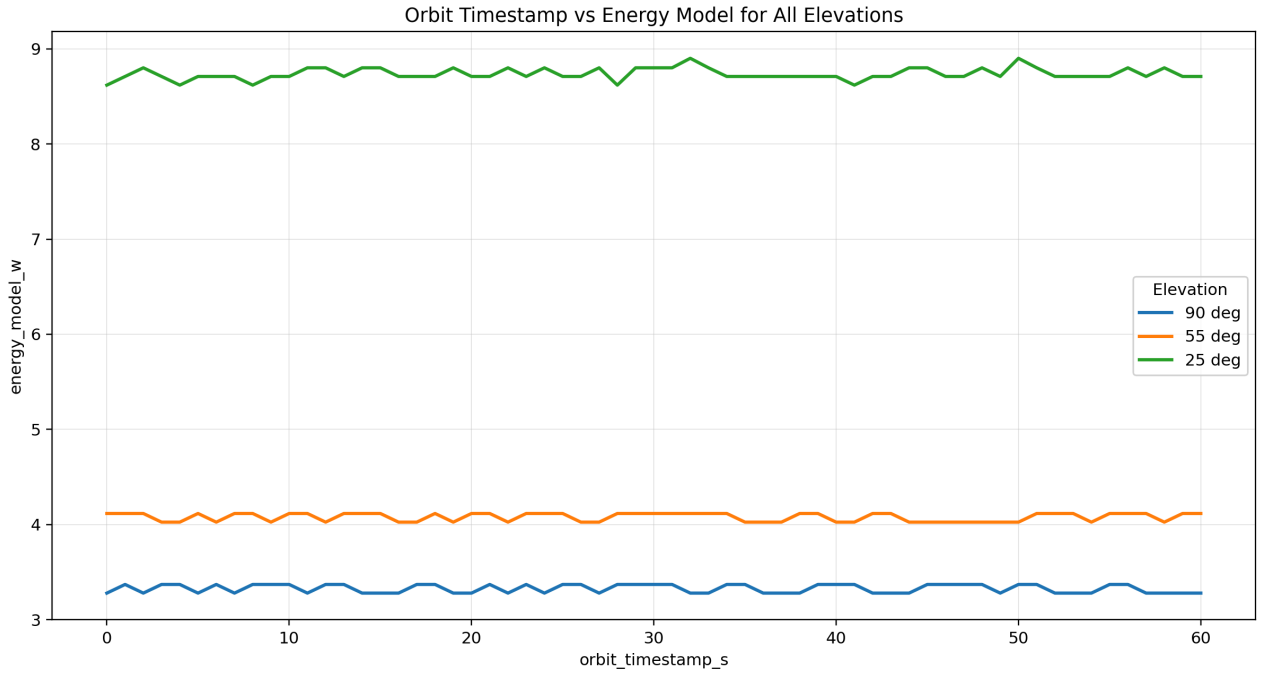


Demodulator State Results

Busy/Idle vs Orbit Timestamp (25 deg)



Energy Model Results (90, 55 and 25 deg)



- Energy follows geometry-driven demod state transitions.

Future Work

Goal: move from scenario simulation to validated, adaptive NTN system design.

1. Validate with real data: calibrate traffic, channel loss, and demod behavior using measurements.
2. Add uncertainty modeling: include weather, bursty traffic, and mobility distributions with confidence intervals.
3. Compare access schemes: benchmark LR-FHSS against alternative PHY/MAC options under equal constraints.

Future Work

4. Optimize control: design dynamic demod/power allocation policies using optimization or reinforcement learning.
5. Scale to network level: extend from single-satellite view to multi-satellite and inter-satellite coordination.
6. Build reproducible benchmarks: publish datasets, metrics, and open evaluation protocols for fair comparison.

Citable Paper Sources (URLs)

- LR-FHSS overview paper: https://doi.org/10.1109/MCOM.001.2000627
- 3GPP NTN reference (TR 38.811): https://www.3gpp.org/DynaReport/38.811.htm
- Friis transmission formula: https://doi.org/10.1109/JRPROC.1946.234568
- Shannon communication theory: https://doi.org/10.1002/j.1538-7305.1948.tb01338.x
- ALOHA system paper: https://doi.org/10.1145/1478462.1478502

Citable Data Sources (URLs)

- Natural Earth populated places: https://naciscdn.org/naturalearth/10m/cultural/ne_10m_populated_places.zip
- Natural Earth countries: https://naciscdn.org/naturalearth/10m/cultural/ne_10m_admin_0_countries.zip
- Natural Earth lakes: https://naciscdn.org/naturalearth/10m/physical/ne_10m_lakes.zip
- Natural Earth rivers: https://naciscdn.org/naturalearth/10m/physical/ne_10m_rivers_lake_centerlines.zip
- Demod power baseline reference: https://tnm.engin.umich.edu/wp-content/uploads/sites/353/2017/12/2006.10.Reducing-idle-mode-power-in-software-defined-radio-terminals_ISLPED-2006.pdf